Nexus Platform - Database Design and Data Modeling Blueprint

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SECTION 1: RELATIONAL DATABASE SCHEMA (POSTGRESQL 16)

The primary transactional database model includes the following core relational schemas:

- ‘users‘: Stores user identity, hashed credentials, role (‘admin‘, ‘editor‘, ‘viewer‘), and assigned department (‘general‘, ‘engineering‘, ‘finance‘, ‘hr‘, ‘legal‘).

- ‘conversations‘: Primary chat threads identified by UUIDv7, storing ‘user_id‘, ‘title‘, ‘created_at‘, and ‘updated_at‘.

- ‘chat_messages‘: Stores message records with ‘conversation_id‘, ‘role‘ (‘user‘, ‘assistant‘), ‘content‘, and JSONB ‘citations‘ metadata.

- ‘audit_logs‘: Partitioned monthly by timestamp to record all access attempts, query text, and security authorization checks.

SECTION 2: VECTOR DATABASE SCHEMA (QDRANT)

The Qdrant collection ‘hybrid_knowledge_base‘ is structured with dual vector spaces:

- Dense Vector Space: 384 dimensions using Cosine distance, generated by ‘BAAI/bge-small-en-v1.5‘ for semantic embeddings.

- Sparse Vector Space: SPLADE lexical weights (‘prithivida/Splade_PP_en_v1‘) for exact term and keyword recall.

- Payload Keyword Indices: Dedicated payload indices on ‘department‘ (enabling sub-millisecond RBAC filtering) and ‘source_file‘.

SECTION 3: REPLICATION, BACKUP, AND DISASTER RECOVERY

The PostgreSQL database runs with a Multi-AZ standby replica with synchronous replication. Point-in-time recovery (PITR) is maintained through continuous Write-Ahead Log (WAL) archiving to encrypted Amazon S3 buckets. Automated full snapshots execute daily at 02:00 UTC and are retained for 35 days in compliance with enterprise audit criteria.